

Python Interview Practice Questions

Easy to Medium | Arrays · Lists · Strings · Dicts · Sets · Sorting · Time Complexity

Topic	Easy	Medium
1. Arrays & Lists	20 Easy	15 Medium
2. Strings		15 Medium
3. Dictionaries		10 Medium
4. Sets	10 Easy	8 Medium
5. Sorting & Searching	10 Easy	10 Medium
6. Time & Space Complexity		10 Medium
7. Mixed / Real Interview Style		10 Medium
TOTAL	95 Easy	78 Medium

How to use this sheet

- Solve each problem on paper or IDE before running it.
- For easy problems: aim to solve in under 10 minutes.
- For medium problems: aim to solve in 20-30 minutes. If stuck, think about which data structure helps.
- After solving, always ask yourself: Can I improve the time complexity? Can I reduce space?
- Mark each question as: Not Attempted / Attempted / Solved / Optimized.
- Revisit all Attempted-but-not-solved questions after 2 days.

1. Arrays & Lists

EASY

1. Q1. Find the largest and smallest element in a list without using built-in min/max.
2. Q2. Count how many times a given element appears in a list.
3. Q3. Check if a list is sorted in ascending order.
4. Q4. Find the sum and average of all elements in a list.
5. Q5. Remove all occurrences of a specific value from a list.
6. Q6. Reverse a list without using the reverse() method or slicing.
7. Q7. Find the second largest element in a list without sorting.
8. Q8. Return all elements that appear more than once in a list.
9. Q9. Flatten a list of lists into a single list. e.g. `[[1,2],[3,4]] → [1,2,3,4]`
10. Q10. Given a list of numbers, return a new list with each element squared.
11. Q11. Find the index of the first occurrence of a given value.
12. Q12. Swap the first and last elements of a list.
13. Q13. Remove all even numbers from a list.
14. Q14. Return the middle element of a list. If even length, return both middle elements.
15. Q15. Check if two lists have any elements in common.
16. Q16. Given a list, move all zeros to the end while preserving the order of other elements.
17. Q17. Find the difference between the maximum and minimum values in a list.
18. Q18. Return elements that are present in the first list but not in the second.
19. Q19. Merge two lists into a list of tuples. e.g. `[1,2],[a,b] → [(1,a),(2,b)]`
20. Q20. Given a list, return True if all elements are unique.

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21. Q21. Two Sum: Given a list and a target, return indices of two numbers that add up to the target. Solve in $O(n)$.
22. Q22. Rotate a list by k positions to the right. e.g. `[1,2,3,4,5], k=2 → [4,5,1,2,3]`
23. Q23. Find all pairs in a list whose sum equals a given k.
24. Q24. Maximum subarray sum (Kadane's Algorithm). e.g. `[-2,1,-3,4,-1,2,1,-5,4] → 6`
25. Q25. Merge two sorted lists without using sort(). Result must also be sorted.
26. Q26. Given a list of integers, find the length of the longest consecutive sequence. e.g. `[100,4,200,1,3,2] → 4`
27. Q27. Find the majority element (appears more than $n/2$ times) in a list.
28. Q28. Given a list of 0s, 1s, and 2s, sort it in one pass without using sort(). (Dutch National Flag)
29. Q29. Product of Array Except Self: return a list where each element is the product of all other elements. No division allowed.
30. Q30. Find all unique triplets in a list that sum to zero.
31. Q31. Given a list of intervals, merge all overlapping intervals. e.g. `[[1,3],[2,6],[8,10]] → [[1,6],[8,10]]`

- 32. Q32. Find the minimum number of jumps to reach the last index from the first. Each element is the max jump length.
 - 33. Q33. Given a list, find the contiguous subarray with the maximum product.
 - 34. Q34. Rearrange a list so positives and negatives alternate. Extra elements go at the end.
 - 35. Q35. Find k largest elements from a list without fully sorting it.
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2. Strings

EASY

- 36. Q36. Check if a string is a palindrome (ignore spaces and case).
- 37. Q37. Count the frequency of each character in a string.
- 38. Q38. Reverse the words in a sentence. e.g. 'hello world' → 'world hello'
- 39. Q39. Check if two strings are anagrams of each other.
- 40. Q40. Count the number of vowels and consonants in a string.
- 41. Q41. Remove all whitespace from a string.
- 42. Q42. Check if a string contains only digits.
- 43. Q43. Find the first non-repeating character in a string.
- 44. Q44. Convert a string to title case without using .title().
- 45. Q45. Count how many words are in a sentence.
- 46. Q46. Check if a string starts and ends with the same character.
- 47. Q47. Replace all spaces in a string with underscores.
- 48. Q48. Return the most frequent character in a string.
- 49. Q49. Check if one string is a rotation of another. e.g. 'abcde' and 'cdeab'
- 50. Q50. Remove duplicate characters from a string, preserving order.
- 51. Q51. Find all characters that appear in both strings.
- 52. Q52. Given a string, print all its characters in reverse order using a loop.
- 53. Q53. Check if a string is a pangram (contains every letter of the alphabet).
- 54. Q54. Convert a sentence to pig latin (move first consonant to end, add 'ay').
- 55. Q55. Count occurrences of a substring within a string without using count().

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- 56. Q56. Find the longest substring without repeating characters. e.g. 'abcabcbb' → 3
- 57. Q57. String compression: compress consecutive characters. e.g. 'aaabbbccdddee' → 'a3b3c2d4e2'
- 58. Q58. Given a string with brackets, check if they are balanced. e.g. '({[]})' → True
- 59. Q59. Find all permutations of a given string.
- 60. Q60. Find the longest common prefix among a list of strings.
- 61. Q61. Given a string, return all substrings that are palindromes.
- 62. Q62. Implement strStr(): find the first occurrence of needle in haystack without using find().
- 63. Q63. Given a string of words, reverse each word individually but keep word order.
- 64. Q64. Find the minimum window substring that contains all characters of another string.
- 65. Q65. Given a string, find the longest palindromic substring.
- 66. Q66. Group anagrams together from a list of strings. e.g. ['eat','tea','tan','ate','nat','bat'] → [['eat','tea','ate'], ['tan','nat'], ['bat']]
- 67. Q67. Given a string, find if it follows a given pattern. e.g. pattern='abba', str='dog cat cat dog' → True
- 68. Q68. Count the number of distinct substrings of a given string.

69. Q69. Given a string with Roman numerals, convert it to an integer.

70. Q70. Implement wildcard pattern matching where '?' matches any single character and '*' matches any sequence.

3. Dictionaries

EASY

- 71. Q71. Count the frequency of each word in a sentence using a dictionary.
- 72. Q72. Invert a dictionary so keys become values and values become keys.
- 73. Q73. Merge two dictionaries. If duplicate keys exist, keep the value from the second dict.
- 74. Q74. Find the key with the maximum value in a dictionary.
- 75. Q75. Check if two dictionaries have the same keys.
- 76. Q76. Remove a key from a dictionary if it exists, without raising an error.
- 77. Q77. Sort a dictionary by its values in descending order.
- 78. Q78. Group a list of words by their first letter using a dictionary.
- 79. Q79. Given a list of tuples, convert it into a dictionary.
- 80. Q80. Find all keys in a dictionary whose value is greater than a given threshold.
- 81. Q81. Create a dictionary from two lists (one for keys, one for values).
- 82. Q82. Count how many keys in a dict have a value equal to a given number.
- 83. Q83. Return the dictionary with only keys that are present in a given list.
- 84. Q84. Given a nested dictionary, access a deeply nested value safely.
- 85. Q85. Find common keys between two dictionaries.

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- 86. Q86. Find the top k frequent elements in a list using a dictionary.
 - 87. Q87. Given two lists, find common elements with their frequencies as a dictionary.
 - 88. Q88. Word frequency map: given a paragraph, return a dict sorted by word frequency (most frequent first).
 - 89. Q89. Given a list of dicts, sort by multiple keys. e.g. sort by age asc, then name asc.
 - 90. Q90. Flatten a nested dictionary into a single-level dict with dot-notation keys. e.g. `{'a':{'b':1}}` → `{'a.b':1}`
 - 91. Q91. Given a list of student records (dicts), group them by grade using a dictionary.
 - 92. Q92. Two Sum with dictionary: return all pairs (not just one) using a hashmap approach.
 - 93. Q93. Given a sentence, find all words that appear exactly k times.
 - 94. Q94. Design a simple in-memory key-value store with set, get, delete, and get_all operations.
 - 95. Q95. LRU Cache: implement a simple cache that evicts the least recently used item when full (use dict + list or OrderedDict).
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4. Sets

EASY

- 96. Q96. Find the union, intersection, and difference of two lists using sets.
- 97. Q97. Remove duplicates from a list using a set (order doesn't matter).
- 98. Q98. Check if one list is a subset of another.
- 99. Q99. Check if two lists are disjoint (no common elements).
- 100. Q100. Find elements that are in exactly one of two lists (symmetric difference).
- 101. Q101. Given a list, find all unique elements and return them sorted.
- 102. Q102. Check if a string has all unique characters using a set.
- 103. Q103. Find the missing number in a list from 1 to n using a set.
- 104. Q104. Given two sets, find elements that are in the first set but not the second.
- 105. Q105. Count the number of distinct elements in a list.

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- 106. Q106. Given a list of integers, find all pairs with a difference of k using a set.
 - 107. Q107. Find the longest consecutive subsequence in an unsorted list using a set. e.g.
[100,4,200,1,3,2] → 4
 - 108. Q108. Given a list of words, find all words that are subsets of characters of a given string.
 - 109. Q109. Happy Number: a number is happy if repeatedly replacing it with the sum of squares of its digits eventually reaches 1. Detect cycle using a set.
 - 110. Q110. Given a list of emails, find duplicate email addresses using a set.
 - 111. Q111. Find the first duplicate element in a list using a set (preserve original order).
 - 112. Q112. Set cover approximation: given a universe set and a collection of sets, find minimum sets to cover the universe (greedy approach).
 - 113. Q113. Given a list of words, group them into sets of words that are anagrams of each other.
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5. Sorting & Searching

EASY

- 114. Q114. Implement bubble sort on a list.
- 115. Q115. Implement selection sort on a list.
- 116. Q116. Implement insertion sort on a list.
- 117. Q117. Binary search: find target in a sorted list. Return index or -1.
- 118. Q118. Sort a list of tuples by the second element.
- 119. Q119. Sort a list of strings by their length.
- 120. Q120. Sort a list of dictionaries by a given key.
- 121. Q121. Find the index of the first element greater than a given value in a sorted list.
- 122. Q122. Sort a list of mixed positive and negative numbers so negatives come first.
- 123. Q123. Check if a list is sorted using a single loop.

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- 124. Q124. Implement merge sort.
 - 125. Q125. Implement quick sort.
 - 126. Q126. Find the kth smallest element in an unsorted list without fully sorting it.
 - 127. Q127. Given a sorted list that has been rotated, find the minimum element.
 - 128. Q128. Search in a rotated sorted list (binary search variant).
 - 129. Q129. Sort a list of strings such that all anagrams are grouped together.
 - 130. Q130. Given a list of intervals sorted by start time, insert a new interval and merge if necessary.
 - 131. Q131. Find peak element: an element greater than its neighbors. Return its index.
 - 132. Q132. Given a 2D matrix where rows and columns are sorted, search for a target efficiently.
 - 133. Q133. Given n sorted lists, merge them into one sorted list efficiently.
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6. Time & Space Complexity

EASY

- 134. Q134. What is the time complexity of accessing an element by index in a Python list?
- 135. Q135. What is the time complexity of the 'in' operator for a list vs a set vs a dict?
- 136. Q136. What is the time complexity of appending to a list vs inserting at the beginning?
- 137. Q137. What is the time complexity of Python's built-in `sort()` and `sorted()`?
- 138. Q138. Explain the difference between $O(1)$, $O(n)$, $O(n \log n)$, and $O(n^2)$ with simple examples.
- 139. Q139. What is the space complexity of creating a new list from an existing list using list comprehension?
- 140. Q140. What is the time complexity of `dict.get()`, `dict.set()`, and `dict.delete()`?
- 141. Q141. Which is faster for lookup: a list with 1 million items or a set with 1 million items? Why?
- 142. Q142. What is the time complexity of concatenating two strings of length m and n in Python?
- 143. Q143. What is the time complexity of removing an element from the middle of a list?

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- 144. Q144. Write a function to find duplicates in a list. Give two implementations — one $O(n^2)$ and one $O(n)$. Explain the tradeoff.
 - 145. Q145. Analyze the time and space complexity of Kadane's algorithm for maximum subarray sum.
 - 146. Q146. What is the time complexity of the Two Sum problem using a brute force approach vs a hashmap approach?
 - 147. Q147. Explain why dict and set lookups are $O(1)$ average but $O(n)$ worst case.
 - 148. Q148. Given a function with a nested loop where the inner loop runs from i to n for each i , what is the total time complexity?
 - 149. Q149. Compare the time complexity of BFS and DFS on a graph with V vertices and E edges.
 - 150. Q150. What is amortized $O(1)$ time? Use Python list's `append()` as an example.
 - 151. Q151. Analyze the space complexity of a recursive function that goes n levels deep.
 - 152. Q152. You have two solutions: one is $O(n \log n)$ time with $O(1)$ space, another is $O(n)$ time with $O(n)$ space. When would you choose each?
 - 153. Q153. Given the code below, identify the time complexity:

```
for i in range(n):  
    for j in range(i, n):  
        print(i, j)
```
-

7. Mixed / Real Interview Style

EASY

- 154. Q154. Given a list of temperatures, find the days where temperature was above average.
- 155. Q155. You are given a list of transactions (positive = credit, negative = debit). Find the running balance after each transaction.
- 156. Q156. Given a list of names, return only names that start with a vowel.
- 157. Q157. A teacher has a dict of {student: score}. Return a list of students who passed (score >= 40).
- 158. Q158. Given a string of comma-separated values, parse it into a list and remove duplicates.
- 159. Q159. Write a function that takes a list of words and returns the longest word.
- 160. Q160. Given a dict of {item: price}, apply a 10% discount to all items above Rs 500.
- 161. Q161. Find the most common word in a paragraph, ignoring common stop words like 'the', 'a', 'is'.
- 162. Q162. Given a list of emails, return only valid emails (contain '@' and '.').
- 163. Q163. You have a list of tuples (name, score). Return the name of the person with the highest score.

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- 164. Q164. Design a function that takes a list of log entries (timestamps + events) and returns the top 3 most frequent events.
 - 165. Q165. Given a list of product dicts with 'name', 'category', and 'price', group them by category and find the avg price per category.
 - 166. Q166. You have an API response as a nested dict/list. Extract all 'id' fields recursively, regardless of nesting level.
 - 167. Q167. Simulate a simple queue using two stacks (lists). Implement enqueue and dequeue.
 - 168. Q168. Given a list of user activity dicts with 'user_id' and 'action', find users who performed all of: login, purchase, logout.
 - 169. Q169. Rate Limiter: implement a function that allows at most 5 calls per 60 seconds per user_id using a dict.
 - 170. Q170. Given a list of words, find the word that can be formed using the most number of distinct characters from a given character set.
 - 171. Q171. You have a list of (start_time, end_time) intervals. Find the maximum number of overlapping intervals at any point in time.
 - 172. Q172. Given a large list of integers, find the top 3 most frequent numbers without sorting the entire list.
 - 173. Q173. Implement a simple in-memory cache with a TTL (time-to-live). Items expire after a set number of seconds.
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Progress Tracker

Mark each question as you practice: [] Not Done [~] Attempted [✓] Solved [★] Optimized

Q#	Topic	Difficulty	Status
1	Find the largest and smallest element in a list without using built-in min/max.	Easy	[]
2	Count how many times a given element appears in a list.	Easy	[]
3	Check if a list is sorted in ascending order.	Easy	[]
4	Find the sum and average of all elements in a list.	Easy	[]
5	Remove all occurrences of a specific value from a list.	Easy	[]
6	Reverse a list without using the reverse() method or slicing.	Easy	[]
7	Find the second largest element in a list without sorting.	Easy	[]
8	Return all elements that appear more than once in a list.	Easy	[]
9	Flatten a list of lists into a single list. e.g. [[1,2],[3,4]] → [1,2,3,4]	Easy	[]
10	Given a list of numbers, return a new list with each element squared.	Easy	[]
11	Find the index of the first occurrence of a given value.	Easy	[]
12	Swap the first and last elements of a list.	Easy	[]
13	Remove all even numbers from a list.	Easy	[]
14	Return the middle element of a list. If even length, return both middle elements...	Easy	[]
15	Check if two lists have any elements in common.	Easy	[]
16	Given a list, move all zeros to the end while preserving the order of other elem...	Easy	[]
17	Find the difference between the maximum and minimum values in a list.	Easy	[]
18	Return elements that are present in the first list but not in the second.	Easy	[]
19	Merge two lists into a list of tuples. e.g. [1,2],[a,b] → [(1,a),(2,b)]	Easy	[]
20	Given a list, return True if all elements are unique.	Easy	[]
21	Two Sum: Given a list and a target, return indices of two numbers that add up to...	Medium	[]
22	Rotate a list by k positions to the right. e.g. [1,2,3,4,5], k=2 → [4,5,1,2,3]	Medium	[]
23	Find all pairs in a list whose sum equals a given k.	Medium	[]
24	Maximum subarray sum (Kadane's Algorithm). e.g. [-2,1,-3,4,-1,2,1,-5,4] → 6	Medium	[]
25	Merge two sorted lists without using sort(). Result must also be sorted.	Medium	[]
26	Given a list of integers, find the length of the longest consecutive sequence. e...	Medium	[]

27	Find the majority element (appears more than $n/2$ times) in a list.	Medium	[]
28	Given a list of 0s, 1s, and 2s, sort it in one pass without using sort(). (Dutch...	Medium	[]
29	Product of Array Except Self: return a list where each element is the product of...	Medium	[]
30	Find all unique triplets in a list that sum to zero.	Medium	[]
31	Given a list of intervals, merge all overlapping intervals. e.g. [[1,3],[2,6],[8...	Medium	[]
32	Find the minimum number of jumps to reach the last index from the first. Each el...	Medium	[]
33	Given a list, find the contiguous subarray with the maximum product.	Medium	[]
34	Rearrange a list so positives and negatives alternate. Extra elements go at the ...	Medium	[]
35	Find k largest elements from a list without fully sorting it.	Medium	[]
36	Check if a string is a palindrome (ignore spaces and case).	Easy	[]
37	Count the frequency of each character in a string.	Easy	[]
38	Reverse the words in a sentence. e.g. 'hello world' → 'world hello'	Easy	[]
39	Check if two strings are anagrams of each other.	Easy	[]
40	Count the number of vowels and consonants in a string.	Easy	[]
41	Remove all whitespace from a string.	Easy	[]
42	Check if a string contains only digits.	Easy	[]
43	Find the first non-repeating character in a string.	Easy	[]
44	Convert a string to title case without using .title().	Easy	[]
45	Count how many words are in a sentence.	Easy	[]
46	Check if a string starts and ends with the same character.	Easy	[]
47	Replace all spaces in a string with underscores.	Easy	[]
48	Return the most frequent character in a string.	Easy	[]
49	Check if one string is a rotation of another. e.g. 'abcde' and 'cdeab'	Easy	[]
50	Remove duplicate characters from a string, preserving order.	Easy	[]
51	Find all characters that appear in both strings.	Easy	[]
52	Given a string, print all its characters in reverse order using a loop.	Easy	[]
53	Check if a string is a pangram (contains every letter of the alphabet).	Easy	[]
54	Convert a sentence to pig latin (move first consonant to end, add 'ay').	Easy	[]
55	Count occurrences of a substring within a string without using count().	Easy	[]

56	Find the longest substring without repeating characters. e.g. 'abcabcbb' → 3	Medium	[]
57	String compression: compress consecutive characters. e.g. 'aaabbbccdddeee' → 'a3...	Medium	[]
58	Given a string with brackets, check if they are balanced. e.g. '({[]})' → True	Medium	[]
59	Find all permutations of a given string.	Medium	[]
60	Find the longest common prefix among a list of strings.	Medium	[]
61	Given a string, return all substrings that are palindromes.	Medium	[]
62	Implement strStr(): find the first occurrence of needle in haystack without usin...	Medium	[]
63	Given a string of words, reverse each word individually but keep word order.	Medium	[]
64	Find the minimum window substring that contains all characters of another string...	Medium	[]
65	Given a string, find the longest palindromic substring.	Medium	[]
66	Group anagrams together from a list of strings. e.g. ['eat', 'tea', 'tan', 'ate', 'n...	Medium	[]
67	Given a string, find if it follows a given pattern. e.g. pattern='abba', str='do...	Medium	[]
68	Count the number of distinct substrings of a given string.	Medium	[]
69	Given a string with Roman numerals, convert it to an integer.	Medium	[]
70	Implement wildcard pattern matching where '?' matches any single character and '...	Medium	[]
71	Count the frequency of each word in a sentence using a dictionary.	Easy	[]
72	Invert a dictionary so keys become values and values become keys.	Easy	[]
73	Merge two dictionaries. If duplicate keys exist, keep the value from the second ...	Easy	[]
74	Find the key with the maximum value in a dictionary.	Easy	[]
75	Check if two dictionaries have the same keys.	Easy	[]
76	Remove a key from a dictionary if it exists, without raising an error.	Easy	[]
77	Sort a dictionary by its values in descending order.	Easy	[]
78	Group a list of words by their first letter using a dictionary.	Easy	[]
79	Given a list of tuples, convert it into a dictionary.	Easy	[]
80	Find all keys in a dictionary whose value is greater than a given threshold.	Easy	[]
81	Create a dictionary from two lists (one for keys, one for values).	Easy	[]
82	Count how many keys in a dict have a value equal to a given number.	Easy	[]
83	Return the dictionary with only keys that are present in a given list.	Easy	[]

84	Given a nested dictionary, access a deeply nested value safely.	Easy	[]
85	Find common keys between two dictionaries.	Easy	[]
86	Find the top k frequent elements in a list using a dictionary.	Medium	[]
87	Given two lists, find common elements with their frequencies as a dictionary.	Medium	[]
88	Word frequency map: given a paragraph, return a dict sorted by word frequency (m...	Medium	[]
89	Given a list of dicts, sort by multiple keys. e.g. sort by age asc, then name as...	Medium	[]
90	Flatten a nested dictionary into a single-level dict with dot-notation keys. e.g...	Medium	[]
91	Given a list of student records (dicts), group them by grade using a dictionary.	Medium	[]
92	Two Sum with dictionary: return all pairs (not just one) using a hashmap approach...	Medium	[]
93	Given a sentence, find all words that appear exactly k times.	Medium	[]
94	Design a simple in-memory key-value store with set, get, delete, and get_all operations...	Medium	[]
95	LRU Cache: implement a simple cache that evicts the least recently used item when...	Medium	[]
96	Find the union, intersection, and difference of two lists using sets.	Easy	[]
97	Remove duplicates from a list using a set (order doesn't matter).	Easy	[]
98	Check if one list is a subset of another.	Easy	[]
99	Check if two lists are disjoint (no common elements).	Easy	[]
100	Find elements that are in exactly one of two lists (symmetric difference).	Easy	[]
101	Given a list, find all unique elements and return them sorted.	Easy	[]
102	Check if a string has all unique characters using a set.	Easy	[]
103	Find the missing number in a list from 1 to n using a set.	Easy	[]
104	Given two sets, find elements that are in the first set but not the second.	Easy	[]
105	Count the number of distinct elements in a list.	Easy	[]
106	Given a list of integers, find all pairs with a difference of k using a set.	Medium	[]
107	Find the longest consecutive subsequence in an unsorted list using a set. e.g. [...	Medium	[]
108	Given a list of words, find all words that are subsets of characters of a given ...	Medium	[]
109	Happy Number: a number is happy if repeatedly replacing it with the sum of squares...	Medium	[]
110	Given a list of emails, find duplicate email addresses using a set.	Medium	[]

111	Find the first duplicate element in a list using a set (preserve original order)...	Medium	[]
112	Set cover approximation: given a universe set and a collection of sets, find min...	Medium	[]
113	Given a list of words, group them into sets of words that are anagrams of each o...	Medium	[]
114	Implement bubble sort on a list.	Easy	[]
115	Implement selection sort on a list.	Easy	[]
116	Implement insertion sort on a list.	Easy	[]
117	Binary search: find target in a sorted list. Return index or -1.	Easy	[]
118	Sort a list of tuples by the second element.	Easy	[]
119	Sort a list of strings by their length.	Easy	[]
120	Sort a list of dictionaries by a given key.	Easy	[]
121	Find the index of the first element greater than a given value in a sorted list.	Easy	[]
122	Sort a list of mixed positive and negative numbers so negatives come first.	Easy	[]
123	Check if a list is sorted using a single loop.	Easy	[]
124	Implement merge sort.	Medium	[]
125	Implement quick sort.	Medium	[]
126	Find the kth smallest element in an unsorted list without fully sorting it.	Medium	[]
127	Given a sorted list that has been rotated, find the minimum element.	Medium	[]
128	Search in a rotated sorted list (binary search variant).	Medium	[]
129	Sort a list of strings such that all anagrams are grouped together.	Medium	[]
130	Given a list of intervals sorted by start time, insert a new interval and merge ...	Medium	[]
131	Find peak element: an element greater than its neighbors. Return its index.	Medium	[]
132	Given a 2D matrix where rows and columns are sorted, search for a target efficie...	Medium	[]
133	Given n sorted lists, merge them into one sorted list efficiently.	Medium	[]
134	What is the time complexity of accessing an element by index in a Python list?	Easy	[]
135	What is the time complexity of the 'in' operator for a list vs a set vs a dict?	Easy	[]
136	What is the time complexity of appending to a list vs inserting at the beginning...	Easy	[]
137	What is the time complexity of Python's built-in sort() and sorted()?	Easy	[]
138	Explain the difference between $O(1)$, $O(n)$, $O(n \log n)$, and $O(n^2)$ with simple exa...	Easy	[]

139	What is the space complexity of creating a new list from an existing list using ...	Easy	[]
140	What is the time complexity of dict.get(), dict.set(), and dict.delete()?	Easy	[]
141	Which is faster for lookup: a list with 1 million items or a set with 1 million ...	Easy	[]
142	What is the time complexity of concatenating two strings of length m and n in Py...	Easy	[]
143	What is the time complexity of removing an element from the middle of a list?	Easy	[]
144	Write a function to find duplicates in a list. Give two implementations — one O(...	Medium	[]
145	Analyze the time and space complexity of Kadane's algorithm for maximum subarray...	Medium	[]
146	What is the time complexity of the Two Sum problem using a brute force approach ...	Medium	[]
147	Explain why dict and set lookups are O(1) average but O(n) worst case.	Medium	[]
148	Given a function with a nested loop where the inner loop runs from i to n for ea...	Medium	[]
149	Compare the time complexity of BFS and DFS on a graph with V vertices and E edge...	Medium	[]
150	What is amortized O(1) time? Use Python list's append() as an example.	Medium	[]
151	Analyze the space complexity of a recursive function that goes n levels deep.	Medium	[]
152	You have two solutions: one is O(n log n) time with O(1) space, another is O(n) ...	Medium	[]
153	Given the code below, identify the time complexity: for i in range(n): ...	Medium	[]
154	Given a list of temperatures, find the days where temperature was above average.	Easy	[]
155	You are given a list of transactions (positive = credit, negative = debit). Find...	Easy	[]
156	Given a list of names, return only names that start with a vowel.	Easy	[]
157	A teacher has a dict of {student: score}. Return a list of students who passed (...	Easy	[]
158	Given a string of comma-separated values, parse it into a list and remove duplic...	Easy	[]
159	Write a function that takes a list of words and returns the longest word.	Easy	[]
160	Given a dict of {item: price}, apply a 10% discount to all items above Rs 500.	Easy	[]
161	Find the most common word in a paragraph, ignoring common stop words like 'the',...	Easy	[]
162	Given a list of emails, return only valid emails (contain '@' and '.').	Easy	[]

163	You have a list of tuples (name, score). Return the name of the person with the ...	Easy	[]
164	Design a function that takes a list of log entries (timestamps + events) and ret...	Medium	[]
165	Given a list of product dicts with 'name', 'category', and 'price', group them b...	Medium	[]
166	You have an API response as a nested dict/list. Extract all 'id' fields recursiv...	Medium	[]
167	Simulate a simple queue using two stacks (lists). Implement enqueue and dequeue.	Medium	[]
168	Given a list of user activity dicts with 'user_id' and 'action', find users who ...	Medium	[]
169	Rate Limiter: implement a function that allows at most 5 calls per 60 seconds pe...	Medium	[]
170	Given a list of words, find the word that can be formed using the most number of...	Medium	[]
171	You have a list of (start_time, end_time) intervals. Find the maximum number of ...	Medium	[]
172	Given a large list of integers, find the top 3 most frequent numbers without sor...	Medium	[]
173	Implement a simple in-memory cache with a TTL (time-to-live). Items expire after...	Medium	[]